M Apurva CO1-AT1
192325138

CO1 - AT1 SLOT A
DSA0404-Fundamentals of Data Science

1. What is Data Science? Explain its primary objective.

Answer:
Data Science is an interdisciplinary field that uses scientific methods, algorithms, statistics, and computing techniques to extract meaningful insights and knowledge from structured and unstructured data.

Primary Objective:
The main objective of Data Science is to convert raw data into useful information that supports decision-making, prediction, and problem-solving.

2. Why is Data Science important in today's digital world? Mention any four reasons.

Answer:
Data Science is important because:

1. Helps organizations make data-driven decisions.
2. Enables prediction of future trends and customer behavior.
3. Improves business efficiency and automation.
4. Supports innovation in technologies such as Artificial Intelligence and Machine Learning.

3. List any four benefits of Data Science in business and society.

Answer:

1. Improved decision-making.
2. Increased operational efficiency.
3. Better customer experience and personalization.
4. Enhanced healthcare, education, and public services.

4. Mention four real-world applications of Data Science.

Answer:

1. Healthcare – Disease prediction and medical diagnosis.
2. Finance – Fraud detection and risk analysis.
3. E-commerce – Product recommendation systems.

4. Transportation – Traffic prediction and autonomous vehicles.

5. What are the different facets of data? Briefly explain each.

Answer:

1. **Volume:** Amount of data generated.
2. **Variety:** Different forms of data such as text, images, videos, etc.
3. **Velocity:** Speed at which data is generated and processed.
4. **Veracity:** Quality and reliability of data.
5. **Value:** Useful insights obtained from data.

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Type	Description	Examples
Structured Data	Organized in rows and columns	Database tables, Excel sheets
Semi-Structured Data	Has partial organization using tags	XML files, JSON files
Unstructured Data	No predefined format	Images, videos, emails, social media posts

7. What is Big Data? State its key characteristics (5 Vs).

Answer:
Big Data refers to extremely large and complex datasets that cannot be processed using traditional data-processing tools.

5 Vs of Big Data:

1. **Volume** – Large quantity of data.
2. **Velocity** – High speed of data generation.
3. **Variety** – Different types of data.
4. **Veracity** – Data quality and accuracy.

5. **Value** – Useful information extracted from data.

8. Explain the Big Data ecosystem. Name any four technologies used in it.

Answer:
The Big Data ecosystem consists of tools and technologies used for data collection, storage, processing, analysis, and visualization.

Technologies:

1. Hadoop
2. Apache Spark
3. Hive
4. MongoDB

Other technologies include Kafka, HBase, Cassandra, and Flume.

9. What are the major phases of the Data Science process?

Answer:

1. Problem Definition
2. Data Collection
3. Data Cleaning
4. Data Integration
5. Data Transformation
6. Data Analysis
7. Model Building
8. Model Evaluation
9. Deployment
10. Presentation of Results

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Answer:
Data retrieval is the process of obtaining data from various sources for analysis.

Common Sources:

1. Databases (SQL, Oracle, MySQL)
2. Web sources/APIs and Social Media Platforms

---

11. Why is data collection considered an important step in the Data Science process?

Answer:
Data collection is important because the quality of analysis depends on the quality of collected data.

Importance:

1. Provides relevant information for analysis.
2. Improves model accuracy.
3. Reduces bias and errors.
4. Helps in better decision-making.

12. What is data cleansing? List any four common data-cleaning techniques.

Answer:
Data cleansing is the process of detecting and correcting inaccurate, incomplete, or inconsistent data.

Techniques:

1. Handling missing values.
2. Removing duplicate records.
3. Correcting inconsistent data formats.
4. Detecting and removing outliers.

13. What is data integration? Why is it necessary in Data Science?

Answer:
Data integration is the process of combining data from multiple sources into a unified view.

Necessity:

1. Provides complete information.
2. Improves data consistency.
3. Eliminates redundancy.
4. Enhances analytical accuracy.

14. Explain data transformation with suitable examples.

Answer:
Data transformation is the process of converting data into a suitable format for analysis.

Examples:

- Converting dates from **DD/MM/YYYY** to **YYYY-MM-DD**.
- Normalizing values between 0 and 1.
- Encoding categorical data such as converting **Male/Female** into **0/1**.

15. What is data analysis? Mention any four techniques used for data analysis.

Answer:
Data analysis is the process of examining, organizing, and interpreting data to discover useful information and patterns.

Techniques:

1. Statistical Analysis
2. Regression Analysis
3. Classification
4. Clustering

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Analytics Type	Purpose	Example
Descriptive Analytics	Explains what happened	Monthly sales report

Analytics Type	Purpose	Example
Predictive Analytics	Predicts future events	Sales forecasting
Prescriptive Analytics	Suggests actions to take	Product recommendation systems

17. What is meant by building a model in Data Science? Mention the basic steps involved.

Answer:
Model building is the process of creating algorithms that learn patterns from data to make predictions or decisions.

Steps:

1. Select algorithm.
2. Split data into training and testing sets.
3. Train the model.
4. Test the model.
5. Tune model parameters.

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Answer:
Model evaluation helps determine how well a model performs and whether it can be used in real-world applications.

Classification Metrics:

1. Accuracy
2. Precision
3. Recall
4. F1-Score

Other metrics: ROC-AUC, Specificity.

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Answer:
Presenting findings helps stakeholders understand insights and make informed decisions.

Importance:

1. Simplifies complex data.
2. Improves communication.
3. Supports decision-making.
4. Identifies trends and patterns.

Visualization Tools/Techniques:

1. Tableau
2. Microsoft Power BI
3. Matplotlib
4. Seaborn

Techniques: Bar charts, Pie charts, Histograms, Scatter plots.

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

Answer:
After evaluation, models are deployed as web applications, APIs, mobile applications, or cloud services so that users can access predictions in real time.

Examples of Data Science Applications:

1. Recommendation systems in companies like Netflix and Amazon.
2. Fraud detection in banking systems.
3. Healthcare diagnosis and disease prediction.
4. Autonomous vehicles and traffic management systems.

Deployment methods include:

- Cloud platforms
- REST APIs
- Mobile applications
- Embedded systems and IoT devices.